

④ (i) Find the values of k for which the equations $x+y+z=1$,
 $x+2y+4z=k$ & $x+4y+10z=k^2$ may have a solution.
 For these values of k , find the solution also.

Solⁿ:
 Given that,
 $x+y+z=1$
 $x+2y+4z=k$
 $x+4y+10z=k^2$

Step 1:
 The matrix representation of above system is given by

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 10 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ k \\ k^2 \end{bmatrix}$$

Step 2: Rank of Augmented matrix:

$$[A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & k \\ 1 & 4 & 10 & k^2 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & k-1 \\ 0 & 3 & 9 & k^2-1 \end{array} \right] R_3 \rightarrow R_3 - 3R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & k-1 \\ 0 & 0 & 0 & k^2-3k+2 \end{array} \right] \text{ which is the req Echelon form.}$$

Here, we infer that, $R[A] = 2$ & Given that, this system has solution.
 Hence, by Rouché's Theorem, $R[A|B] = 2$, which is possible

only if $k^2 - 3k + 2 = 0$

$$\Rightarrow (k-2)(k-1) = 0$$

$$\Rightarrow k = 1, 2.$$

(ie) This system has solution in two cases.

Case (i) $k = 1$

From the Echelon form

$$x + y + z = 1$$

$$y + 3z = 0$$

Let $\boxed{z = \lambda}$, then $\boxed{y = -3\lambda}$

$$x = 1 + 3\lambda - \lambda$$

$$\therefore \boxed{x = 1 + 2\lambda}$$

Hence, The one-parameter family of solution is

$$\boxed{x = 1 + 2\lambda, y = -3\lambda, z = \lambda}$$

Case (ii) $k = 2$

From the Echelon form,

$$x + y + z = 1$$

$$y + 3z = 1$$

Let $\boxed{z = \mu}$, then $\boxed{y = 1 - 3\mu}$

$$x = 1 - 1 + 3\mu - \mu$$

$$\boxed{x = 2\mu}$$

Hence, the one-parameter family of solution is

$$\boxed{x = 2\mu, y = 1 - 3\mu, z = \mu}$$

(ii) Find the values of λ for which the equations $x + 2y + z = 4$

$2x - y - z = 3\lambda$ & $4x - 7y - 5z = \lambda^2$ may have a solution. For these values of λ find the solution also.

Sol. Given that $x + 2y + z = 4$
 $2x - y - z = 3\lambda$
 $4x - 7y - 5z = \lambda^2$

Step 1:- The matrix representation of above system is given by 10.

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & -1 \\ 4 & -7 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 3\lambda \\ \lambda^2 \end{bmatrix}$$

Step 2:- Rank of Augmented matrix -

$$[A|B] = \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 2 & -1 & -1 & 3\lambda \\ 4 & -7 & -5 & \lambda^2 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 4R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & -5 & -3 & 3\lambda - 8 \\ 0 & -15 & -9 & \lambda^2 - 16 \end{array} \right] R_3 \rightarrow R_3 - 3R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & -5 & -3 & 3\lambda - 8 \\ 0 & 0 & 0 & \lambda^2 - 9\lambda + 8 \end{array} \right]$$

which is the row Echelon form.

Here, we infer that $R[A] = 2$ & Given that, this system may have solution & Hence by Rouché's Theorem, $R[A|B] = 2$, which is possible

$$\text{only if } \lambda^2 - 9\lambda + 8 = 0$$

$$\Rightarrow (\lambda - 8)(\lambda - 1) = 0$$

$$\Rightarrow \lambda = 1, 8.$$

i.e. This system has solution in 2-cases

Case (i) $\lambda = 1$

From Echelon form we have

$$x + 2y + z = 4$$

$$-5y - 3z = -5$$

Let $z = \alpha$, then $-5y = -5 + 3\alpha$

$$y = \frac{3\alpha - 5}{-5} \Rightarrow \frac{5 - 3\alpha}{5} = y$$

$$x = 4 - 2 \left[\frac{5 - 3\alpha}{5} \right] - \alpha$$

$$x = \frac{20 - 10 + 6\alpha - 5\alpha}{5}$$

$$x = \frac{10 + \alpha}{5}$$

Hence, the one-parameter family of solution is

$$x = \frac{10+\alpha}{5}, y = \frac{5-3\alpha}{5}, z = \alpha$$

Case (ii) $\lambda = 8$.

From the Echelon form,

$$x + 2y + z = 4$$

$$-5y - 3z = 16$$

Let $z = \beta$, then $-5y = 16 + 3\beta$

$$y = -\frac{(16+3\beta)}{5}$$

$$x = 4 + 2\left[\frac{16+3\beta}{5}\right] + \beta$$

$$x = \frac{20 + 32 + 6\beta + 5\beta}{5}$$

$$x = \frac{52 + 11\beta}{5} = 16 + \frac{11\beta}{5}$$

Hence, the one-parameter family of solution is

$$x = \frac{52+11\beta}{5}, y = -\frac{(16+3\beta)}{5}, z = \beta$$

- ⑤ (i) Find the values of k for which the equations $kx + y + z = 1$, $x + ky + z = 1$, $x + y + kz = 1$, have (a) infinitely many solutions (b) no solution (c) unique solution.

Given that,

$$kx + y + z = 1$$

$$x + ky + z = 1$$

$$x + y + kz = 1$$

Step 1: The matrix aug is given by

$$\begin{bmatrix} k & 1 & 1 \\ 1 & k & 1 \\ 1 & 1 & k \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Step 2: Rank of Augmented matrix :-

11.

$$[A|B] = \left[\begin{array}{ccc|c} k & 1 & 1 & 1 \\ 1 & k & 1 & 1 \\ 1 & 1 & k & 1 \end{array} \right] R_2 \leftrightarrow R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & k & 1 & 1 \\ k & 1 & 1 & 1 \\ 1 & 1 & k & 1 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - kR_1 \\ R_3 \rightarrow R_3 - R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & k & 1 & 1 \\ 0 & 1-k^2 & 1-k & 1-k \\ 0 & 1-k & k-1 & 0 \end{array} \right] R_3 \rightarrow (1+k)R_3 - R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & k & 1 & 1 \\ 0 & 1-k^2 & 1-k & 1-k \\ 0 & 0 & (k+2)(k-1) & k-1 \end{array} \right]$$

which is the req. Echelon form.

$$\left[\begin{array}{ccc|c} k & 1 & 1 & 1 \\ -k & -k^2 & -k & -k \\ \hline 0 & 1-k^2 & 1-k & 1-k \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 1 & k & 1 \\ -1 & -k & -1 & -1 \\ \hline 0 & 1-k & k-1 & 0 \end{array} \right]$$

$$\left[\begin{array}{cc|c} 1-k^2 & 1-k^2-1 & 0 \\ -(1-k^2) & -(1-k) & -(1-k) \\ \hline 0 & k^2+k-2 & k-1 \end{array} \right]$$

↓
(k+2)(k-1)

Case (i)

If $k=1$, then $R[A]=R[A|B]=1 < 3$

∴ The system has infinite No of solution.

Case (ii)

If $k=-2$, then $R[A]=2 \neq 3=R[A|B]$,

∴ The system has no solution.

Case (iii)

If $k \neq 1, k \neq -2$, then $R[A]=R[A|B]=3$,

∴ The system has unique solution.

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- (ii) Find the values of k for which the equations $kx - 2y + z = 1$,
 $x - 2ky + z = -2$ & $x - 2y + kz = 1$, have (a) no solution.
 (b) many solutions (c) unique solution.

Given that

$$kx - 2y + z = 1$$

$$x - 2ky + z = -2$$

$$x - 2y + kz = 1$$

Step 1:-

The matrix Ref. is

$$\begin{bmatrix} k & -2 & 1 \\ 1 & -2k & 1 \\ 1 & -2 & k \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

Step 2:- Rank of Augmented matrix:-

$$[A|B] = \left[\begin{array}{ccc|c} k & -2 & 1 & 1 \\ 1 & -2k & 1 & -2 \\ 1 & -2 & k & 1 \end{array} \right] R_2 \leftrightarrow R_1$$

$$\begin{array}{ccc|c} k & -2 & 1 & 1 \\ -k & +2k^2 - k & +2k & +2k \\ \hline 0 & -2+2k^2 & 1-k & 1+2k \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & -2k & 1 & -2 \\ k & -2 & 1 & 1 \\ 1 & -2 & k & 1 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - kR_1 \\ R_3 \rightarrow R_3 - R_1 \end{array}$$

$$\begin{array}{ccc|c} 1 & -2 & k & 1 \\ -1 & +2k-1 & -1 & +2 \\ \hline 0 & -2+2k & k-1 & 3 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & -2k & 1 & -2 \\ 0 & 2k^2-2 & 1-k & 1+2k \\ 0 & 2k-2 & k-1 & 3 \end{array} \right] \begin{array}{l} R_2 \rightarrow \frac{R_2}{2} \\ R_3 \rightarrow \frac{R_3}{2} \end{array}$$

$$\begin{array}{ccc|c} k^2-1 & \frac{k^2-1}{2} & \frac{3}{2}(k+1) \\ -(k^2-1) & -\frac{(1-k)}{2} & \frac{(1+2k)}{2} \\ \hline 0 & \frac{1}{2}[k^2-1+k] & \frac{3}{2}k + \frac{3}{2} - \frac{1}{2} - k \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & -2k & 1 & -2 \\ 0 & k^2-1 & \frac{1-k}{2} & \frac{1+2k}{2} \\ 0 & \frac{k-1}{2} & \frac{1-1}{2} & \frac{3}{2} \end{array} \right] R_3 \rightarrow (k+1)R_3 - R_2$$

$$\begin{array}{ccc|c} 0 & \frac{1}{2}[k^2-1+k] & \frac{3}{2}k + \frac{3}{2} - \frac{1}{2} - k \\ 0 & \frac{1}{2}[k^2+k-2] & \frac{k}{2} + 1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & -2k & 1 & -2 \\ 0 & k^2-1 & \frac{1-k}{2} & \frac{1}{2}+k \\ 0 & 0 & \frac{1}{2}[k^2+k-2] & 1+\frac{k}{2} \end{array} \right] R_3 \rightarrow 2R_3$$

$$\downarrow$$

$$(k+2)(k-1)$$

$$\sim \left[\begin{array}{ccc|c} 1 & -2k & 1 & -2 \\ 0 & k^2-1 & \frac{1-k}{2} & \frac{1}{2}+k \\ 0 & 0 & (1+k)(k-1) & 2+k \end{array} \right]$$

(b) If $k = -2$, then $R[A] = R[A|B] = 2 < 3$,
 $\therefore$ The system has infinite no of solution.

(a) If $k = 1$, then $R[A] = 1 \neq 3 = R[A|B]$,
 $\therefore$ The system has no solution.

(c) If $k \neq 1, k \neq -2$, then $R[A] = R[A|B] = 3$.
 $\therefore$ The system has unique solution.

5(iii) For what values of λ, μ does the system of equations
 $x+y+2z=2, 2x-y+3z=2, 5x-y+\lambda z=\mu$ have (a) infinite no of
 solution. (b) no solution. (c) unique solution.

Given that,

$$x+y+2z=2$$

$$2x-y+3z=2$$

$$5x-y+\lambda z=\mu$$

Step 1:- The matrix eqn is given by.

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & 3 \\ 5 & -1 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ \mu \end{bmatrix}$$

Step 2:- Rank of Augmented matrix.

$$[A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 2 & 2 \\ 2 & -1 & 3 & 2 \\ 5 & -1 & \lambda & \mu \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 5R_1 \end{array}$$

$$\begin{array}{ccc|c} 2 & -1 & 3 & 2 \\ -2 & -2 & -4 & -4 \\ \hline 0 & -3 & -1 & -2 \end{array}$$

$$\begin{array}{ccc|c} 5 & -1 & \lambda & \mu \\ -2 & -2 & -10 & -10 \\ \hline 0 & -6 & \lambda-10 & \mu-10 \end{array}$$

$$[A|B] \sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & 2 \\ 0 & -3 & -1 & -2 \\ 0 & -6 & \lambda-10 & \mu-10 \end{array} \right] \quad R_3 \rightarrow R_3 - 2R_2 \quad \begin{array}{cc|c} -6 & \lambda-10 & \mu-10 \\ +6 & +2 & +4 \\ \hline 0 & \lambda-8 & \mu-6 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & 2 \\ 0 & -3 & -1 & -2 \\ 0 & 0 & \lambda-8 & \mu-6 \end{array} \right]$$

(a) If $\mu \neq 6, \lambda \neq 8$, then $R[A] = R[A|B] = 3$
 $(\mu = \text{any value}) \therefore$ The system has unique solution

(b) If $\lambda = 8, \mu \neq 6$, then $R[A] = 2 \neq 3 = R[A|B]$
 $\therefore$ The system has no solution.

(c) If $\lambda = 8, \mu = 6$, then $R[A] = R[A|B] = 2 < 3$
 $\therefore$ The system has Infinitely many solutions.

Q) (iv) For what values of λ, μ does the system of equations $x+y+z=6$,
 $x+2y+3z=10$ & $x+2y+\lambda z=\mu$. (a) many solutions (b) no solutions

(c) only one solution

Given that,

$$\begin{aligned} x+y+z &= 6 \\ x+2y+3z &= 10 \\ x+2y+\lambda z &= \mu \end{aligned}$$

Step 1:- The matrix rep of above is

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ \mu \end{bmatrix}$$

Step 2 The Rank of Augmented matrix:-

$$[A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{array} \right] \begin{array}{l} \\ R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array}$$

$$\begin{array}{ccc|c} 1 & 2 & 3 & 10 \\ -1 & -1 & -1 & -6 \\ \hline 0 & 1 & 2 & 4 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda-1 & \mu-6 \end{array} \right] R_3 \rightarrow R_3 - R_2$$

$$\begin{array}{ccc|c} 1 & 2 & \lambda & \mu \\ -1 & -1 & -1 & -6 \\ \hline 0 & 1 & \lambda-1 & \mu-6 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda-3 & \mu-10 \end{array} \right]$$

$$\begin{array}{ccc|c} 1 & \lambda-1 & \mu-6 \\ -1 & -2 & -4 \\ \hline 0 & \lambda-3 & \mu-10 \end{array}$$

(a) If $\lambda=3, \mu=10$, then $R[A] = R[A|B] = 2 < 3$

$\therefore$ The system has Infinitely no of solutions.

(b) If $\lambda=3, \mu \neq 10$, then $R[A] = 2 \neq 3 = R[A|B]$

$\therefore$ The system has no solution

(c) If $\lambda \neq 3, \mu = \text{any value}$, then $R[A] = R[A|B] = 3$.

$\therefore$ The system has unique solution:-

(6) Find the Non-trivial Solution of the following system of equations if it exists.

$$\begin{aligned} \text{(i)} \quad x - y + 2z - 3w &= 0 \\ 3x + 2y - 4z + w &= 0 \\ 5x - 3y + 2z + 6w &= 0 \\ x - 9y + 14z - 2w &= 0 \end{aligned}$$

Step 1:- The matrix eq is

$$\begin{bmatrix} 1 & -1 & 2 & -3 \\ 3 & 2 & -4 & 1 \\ 5 & -3 & 2 & 6 \\ 1 & -9 & 14 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Step 2:- Rank of A :-

$$[A] = \begin{bmatrix} 1 & -1 & 2 & -3 \\ 3 & 2 & -4 & 1 \\ 5 & -3 & 2 & 6 \\ 1 & -9 & 14 & -2 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 - 3R_1 \\ R_3 \rightarrow R_3 - 5R_1 \\ R_4 \rightarrow R_4 - R_1 \end{array}$$

$$\sim \begin{bmatrix} 1 & -1 & 2 & -3 \\ 0 & 5 & -10 & 10 \\ 0 & 2 & -8 & 21 \\ 0 & -8 & 12 & 1 \end{bmatrix} \begin{array}{l} R_2 \rightarrow \frac{R_2}{5} \end{array}$$

$$\sim \begin{bmatrix} 1 & -1 & 2 & -3 \\ 0 & 1 & -2 & 2 \\ 0 & 2 & -8 & 21 \\ 0 & -8 & 21 & 1 \end{bmatrix} \begin{array}{l} R_3 \rightarrow R_3 - 2R_2 \\ R_4 \rightarrow R_4 + 8R_2 \end{array}$$

$$\sim \begin{bmatrix} 1 & -1 & 2 & -3 \\ 0 & 1 & -2 & 2 \\ 0 & 0 & -4 & 17 \\ 0 & 0 & -4 & 17 \end{bmatrix} \begin{array}{l} R_4 \rightarrow R_4 - R_3 \end{array}$$

$$\sim \begin{bmatrix} 1 & -1 & 2 & -3 \\ 0 & 1 & -2 & 2 \\ 0 & 0 & -4 & 17 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ which is the req Echelon form.}$$

From the Echelon form we infer that

$|A|=0 \Rightarrow$ The system has non-trivial solution.

And we have, (From Echelon form)

$$x - y + 2z - 3w = 0$$

$$y - 2z + 2w = 0$$

$$-4z + 17w = 0$$

$$4z = 17w$$

Let $w = 4\alpha$ then

$$4z = 17(4\alpha)$$

$$z = 17\alpha$$

$$y = 2(17\alpha) - 2(4\alpha)$$

$$y = 34\alpha - 8\alpha$$

$$y = 26\alpha$$

$$x = 26\alpha - 2(17\alpha) + 3(4\alpha)$$

$$x = 26\alpha - 34\alpha + 12\alpha$$

$$x = 4\alpha$$

Thus the non-trivial solution is

$$x = 4\alpha, y = 26\alpha, z = 17\alpha, w = 4\alpha$$

(ii) $x + 2y + z + 2w = 0$

$$x + 3y + 2z + 2w = 0$$

$$2x + 4y + 3z + 6w = 0$$

$$3x + 7y + 4z + 6w = 0.$$

Step 1:- The matrix eqn is

$$\begin{bmatrix} 1 & 2 & 1 & 2 \\ 1 & 3 & 2 & 2 \\ 2 & 4 & 3 & 6 \\ 3 & 7 & 4 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} x &= -4k \\ y &= 2k \\ z &= -2k \\ w &= k. \end{aligned}$$

Step 2:- The rank of A:-

$$[A] = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 1 & 3 & 2 & 2 \\ 2 & 4 & 3 & 6 \\ 3 & 7 & 4 & 6 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - 2R_1 \\ R_4 \rightarrow R_4 - 3R_1 \end{array}$$

$$\begin{array}{cccc|c} 2 & 4 & 3 & 6 & \\ -2 & -4 & -2 & -4 & \\ \hline 0 & 0 & 1 & 2 & \end{array}$$

$$\begin{array}{cccc|c} 3 & 7 & 4 & 6 & \\ -3 & -6 & -3 & -6 & \\ \hline 0 & 1 & 1 & 0 & \end{array}$$

$$\sim \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{array}{l} \\ \\ R_4 \rightarrow R_4 - R_2 \end{array}$$

$$\begin{array}{ccc|c} 1 & 1 & 0 & \\ -1 & -1 & -0 & \\ \hline 0 & 0 & 0 & \end{array}$$

$$\sim \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ which is row Echelon form.}$$

From we infer that, $|A| = 0 \Rightarrow$ The system has non-trivial solution.

$$\begin{aligned} \therefore x + 2y + z + 2w &= 0 \\ y + z &= 0 \\ z + 2w &= 0 \end{aligned}$$

$$\begin{aligned} \text{Let } w &= \alpha, \text{ then } z = -2\alpha, \\ y &= 2\alpha \\ x &= -4\alpha + 2\alpha + 2\alpha \\ &= -4\alpha \end{aligned}$$

Here the non-trivial solution of the system is

$$\begin{aligned} x &= -4\alpha, y = 2\alpha, z = -2\alpha, w = \alpha \\ &\text{(it's a one-parameter family of solution)} \end{aligned}$$

7(ii) Find the values of λ for which the equation

(15)

$3x + y - \lambda z = 0$
 $4x - 2y - 3z = 0$
 $2\lambda x + 4y + \lambda z = 0$ have a non-trivial solution. Also, find the solution in each case.

Step 1: The matrix eq is

$$\begin{bmatrix} 3 & 1 & -\lambda \\ 4 & -2 & -3 \\ 2\lambda & 4 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{array}{r} -2 - \frac{4}{3} \\ -6 - 4 \\ \hline 3 \end{array}$$

Step 2 Rank of A

$$[A] = \begin{bmatrix} 3 & 1 & -\lambda \\ 4 & -2 & -3 \\ 2\lambda & 4 & \lambda \end{bmatrix} R_1 \rightarrow \frac{R_1}{3}$$

$$\begin{array}{rrr} 4 & -2 & -3 \\ -4 & -4/3 & +4\lambda/3 \\ \hline 0 & -10/3 & \frac{4\lambda}{3} - 3 \end{array}$$

$$\sim \begin{bmatrix} 1 & 1/3 & -\lambda/3 \\ 4 & -2 & -3 \\ 2\lambda & 4 & \lambda \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 - 4R_1 \\ R_3 \rightarrow R_3 - (2\lambda)R_1 \end{array}$$

$$\begin{array}{rrr} 2\lambda & 4 & \lambda \\ -2\lambda & -2\lambda/3 & +2\lambda^2/3 \\ \hline 0 & 4 - \frac{2\lambda}{3} & \frac{2\lambda^2}{3} + \lambda \\ 120 - 20\lambda & 20\lambda^2 + 30\lambda & \\ \hline -(120 - 20\lambda) & -[8\lambda^2 - 66\lambda + 108] & \\ \hline 0 & 12\lambda^2 + 96\lambda - 108 & \end{array}$$

$$\sim \begin{bmatrix} 1 & 1/3 & -\lambda/3 \\ 0 & -10/3 & \frac{4\lambda - 9}{3} \\ 0 & \frac{12 - 2\lambda}{3} & \frac{2\lambda^2 + 3\lambda}{3} \end{bmatrix} \begin{array}{l} R_2 \rightarrow -\frac{3}{10} R_2 \\ R_3 \rightarrow 3R_3 \end{array}$$

$$\begin{array}{rrr} 2(6-\lambda) & 2\lambda^2 + 3\lambda & \\ -2(4-\lambda) & -10(9-4\lambda) & 2(6-\lambda) \\ \hline 0 & 2\lambda^2 + 3\lambda - 20[54 - 9\lambda] & -24\lambda + 4\lambda^2 \end{array}$$

$$\sim \begin{bmatrix} 1 & 1/3 & -\lambda/3 \\ 0 & 10 & 9 - 4\lambda \\ 0 & 2(6-\lambda) & 2\lambda^2 + 3\lambda \end{bmatrix} R_3 \rightarrow 10R_3 - 2(6-\lambda)R_2$$

$$\begin{array}{rrr} 0 & 2\lambda^2 + 3\lambda - 20[54 - 33\lambda + 4\lambda^2] & \\ 0 & 2\lambda^2 + 3\lambda - 1080 & \end{array}$$

$$\sim \begin{bmatrix} 1 & 1/3 & -\lambda/3 \\ 0 & 10 & 9 - 4\lambda \\ 0 & 0 & 12\lambda^2 + 96\lambda - 1080 \end{bmatrix} R_3 \rightarrow \frac{R_3}{12}$$

$$[A] \sim \begin{bmatrix} 1 & 1/3 & -\lambda/3 \\ 0 & 10 & 9-4\lambda \\ 0 & 0 & \lambda^2+8\lambda-9 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 1/3 & -\lambda/3 \\ 0 & 10 & 9-4\lambda \\ 0 & 0 & (\lambda+9)(\lambda-1) \end{bmatrix} \text{ which is the req Echelon form}$$

W.K.T, Necessary Condition for a system to have Non-trivial Solution is $|A|=0$, which is possible

when ~~only~~ $\lambda = -9, 1$,

Case (i) $\lambda = 1$.

From Echelon form, $x + y/3 - z/3 = 0$

$$10y + 5z = 0$$

Let $\boxed{z = 2\alpha}$

$$10y = -5(2\alpha)$$

$$\boxed{y = -\alpha}$$

$$x = \frac{\alpha}{3} + \frac{2\alpha}{3} = \frac{3\alpha}{3} = \alpha$$

$$\boxed{x = \alpha}$$

$$\boxed{x = \alpha, y = -\alpha, z = 2\alpha} \text{ if } \lambda = 1$$

One-parameter Family of Solution

$$9 - 4(-9)$$

$$9(1+4)$$

$$9(5) = 45$$

Case (ii) $\lambda = -9$.

From Echelon form $x + y/3 + 3z = 0$

$$10y + 45z = 0$$

Let $z = 2\beta$, $10y = -45(2\beta)$

$$y = -9\beta, \quad x = -\frac{(9\beta)}{3} - 3(2\beta)$$

$$x = +3\beta - 6\beta$$

$$x = -3\beta$$

Take $\boxed{x = -3\beta, y = -9\beta, z = 2\beta}$

one-parameter family of solutions.

7) (ii) Find the values of λ for which the equations.

$$x + (\lambda + 4)y + (4\lambda + 2)z = 0$$

$$x + 2(\lambda + 1)y + (3\lambda + 4)z = 0$$

$$2x + 3\lambda y + (3\lambda + 4)z = 0$$

have a non-trivial solution. Also find the solution in each case.

Step 1 :: The matrix eq is

$$\begin{bmatrix} 1 & \lambda + 4 & 4\lambda + 2 \\ 1 & 2\lambda + 2 & 3\lambda + 4 \\ 2 & 3\lambda & 3\lambda + 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Step 2 The Rank of A :-

$$[A] = \begin{bmatrix} 1 & \lambda + 4 & 4\lambda + 2 \\ 1 & 2\lambda + 2 & 3\lambda + 4 \\ 2 & 3\lambda & 3\lambda + 4 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - 2R_1 \end{array}$$

$$-\lambda + 2 = -(\lambda - 2)$$

$$\sim \begin{bmatrix} 1 & \lambda + 4 & 4\lambda + 2 \\ 0 & \lambda - 2 & -\lambda + 2 \\ 0 & \lambda - 8 & -5\lambda \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 \cdot (-1) \\ R_3 \rightarrow (\lambda - 2)R_3 - (\lambda - 8)R_2 \end{array}$$

$$\sim \begin{bmatrix} 1 & \lambda + 4 & 4\lambda + 2 \\ 0 & 1 & -1 \\ 0 & \lambda - 8 & -5\lambda \end{bmatrix} \begin{array}{l} R_3 \rightarrow R_3 - (\lambda - 8)R_2 \end{array}$$

$$\begin{array}{rcl} & \lambda - 8 & -5\lambda \\ & -(\lambda - 8) & +(\lambda - 8) \\ \hline & 0 & \lambda - 8 - 5\lambda \\ & & -8 - 4\lambda \end{array}$$

$$\sim \begin{bmatrix} 1 & \lambda+4 & 4\lambda+2 \\ 0 & 1 & -1 \\ 0 & 0 & -4\lambda-8 \end{bmatrix}$$

The system has Non-trivial solution only if $|A|=0$

$$\Rightarrow -4\lambda-8=0$$

$$\Rightarrow 4\lambda = -8$$

$$\Rightarrow \boxed{\lambda = -2}$$

$$[A] \sim \begin{bmatrix} 1 & \lambda+4 & 4\lambda+2 \\ 0 & \lambda-2 & -(\lambda-2) \\ 0 & 0 & -4(\lambda-2)(\lambda+2) \end{bmatrix} \text{ which is the Echelon form}$$

The system has Non-trivial solution only if $|A|=0$
 $\Rightarrow \lambda = \pm 2$.

Case (i) $\lambda = 2$.

From the Echelon form

$$x + 6y + 10z = 0$$

$$\text{Let } z = \alpha_1, y = \alpha_2 \Rightarrow x = -6\alpha_2 - 10\alpha_1$$

Have $\boxed{x = -10\alpha_1 - 6\alpha_2, y = \alpha_2, z = \alpha_1}$ which is 2-parameter family of solution.

Case (ii) $\lambda = -2$

From the Echelon form

$$x + 2y - 6z = 0$$

$$-4y + 4z = 0$$

$$\text{Let } z = \alpha, y = \alpha, x = -2\alpha + 6\alpha = 4\alpha$$

$$\boxed{x = 4\alpha}$$

Have $\boxed{x = 4\alpha, y = \alpha, z = \alpha}$

which is the one-parameter family of solution.